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Assignment 1 Draft
Due Date: 23 February 2026

23/31

1.

- MODULE and SESSION
  - There is a one-to-many relationship between a Module and a Session.
  - A MODULE can have many Sessions, but a Session can only belong to one Module.
- SESSION and SCHEDULE
  - The relationship between Session and Schedule is a one-to-many relationship.
  - A Session may be Scheduled zero or many times, but each Schedule must belong to exactly one Session
- STUDENT and SCHEDULE
  - The relationship between a Student and Schedule is a one-to-many relationship.
  - A Student must have at least one Schedule, but each Schedule is associated with exactly one Student.

[6]

2. a)

Table	Primary Key	Foreign Key(s)
STUDENT	stuNum	staffNum
STAFF	staffNum	N/A

[7]

2

b)

Table	Yes / No	Explanation
STUDENT	Yes	Each row has a unique identity.
STAFF	Yes	Each row has a unique identity.

Other pk rule for entity integrity?

2

c)

Table	Yes / No	Explanation
STUDENT	No	The staffNum(FK) 17860 points to a nonexistent entry in the STAFF table
STAFF	No	The table doesn't have foreign keys

3

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3. a) π_{stuNum}(STUDENT)

Use Insert Equation in Word.

[4]

RESULTANT TABLE

stuNum
22405887
22365169
22478172
22233435
22692868
22345427

2

b) $\pi_{stuNum, deptID}$(STUDENT |X| STAFF)

RESULTANT TABLE

stuNum	deptID
22365169	2119
22478172	1871
22233435	2119
22345427	2034

2

4. a)

[6]

- A Unary relationship exists when an entity is related to itself. An example in this case would be a SalesPerson managing other SalesPersons. In this relationship, one SalesPerson plays the role of a Manager, and the other one plays that of a Subordinate. ✓✓
- A Binary relationship is an association between two entities. This is depicted by a SalesPerson making a Sale. ✗ −1
- A Ternary relationship is a relationship between ✓ three entities. An example in this instance is, a SalesPerson makes a Sale, a Product is Sold, and a Customer makes a Purchase.

But here we have 4 entities?

2

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b) SALE is a weak entity. The existence of a Sale depends on the Product being sold, the Customer making the purchase, the SalesPerson who assisted with the Sale and it also has primary keys that are derived from its parent entities.

3

Correct - argument not completely clear.

c) If a new saleID attribute is added to SALE, it would still be a weak entity. For an entity to be weak, the following conditions need to be met:

True

- An entity that is existence-dependent.
- An entity that has a Primary Key that is partially or totally derived from the parent entity in the relationship.

Is this still true in this case?

In order to understand the strength of SALE, we first need to outline that it is an associative entity(association class) between the entities SalesPerson and Product. One and only one SalesPerson can sell zero or many products, and one and only one Product can be sold by zero or many SalesPersons (according to the Crow's model provided). This then renders the sale to be completely dependent on those two entities, as well as the Customer who gets involved in a purchase.

1

What are you saying here?